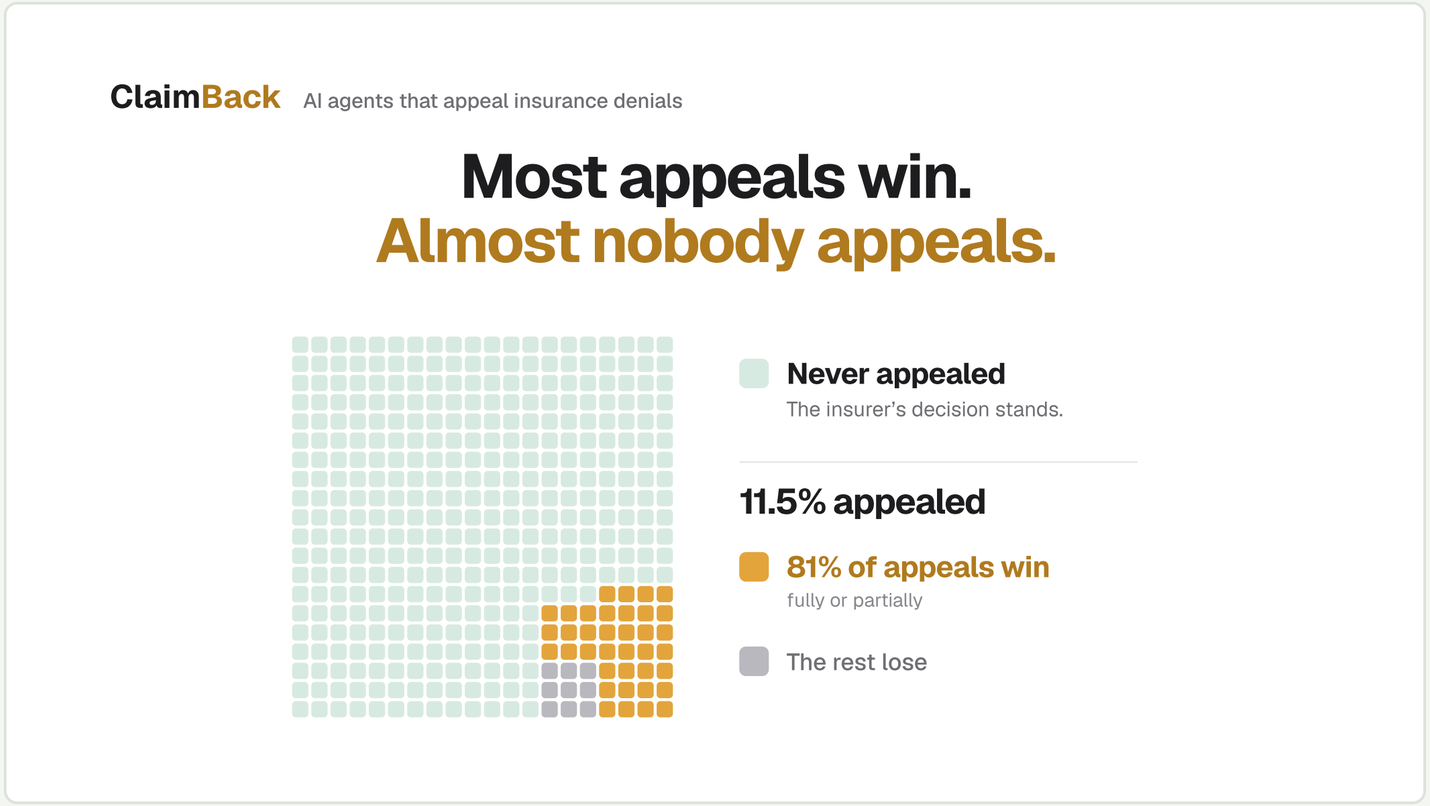


0:00–0:18

Here's the problem. When people appeal an insurance denial, most of them win: eighty-one percent in Medicare Advantage. But almost nobody appeals. Nearly nine in ten denials just stand, and the insurer keeps the money. ClaimBack is a team of AI agents that fights those denials.

BACKGROUND (DON'T READ)

Source line (removed from the slide): KFF: 4.1 million Medicare Advantage prior-authorization denials, 2024 data (January 2026). Each square ≈ 10,250 denials. KFF, "Medicare Advantage Insurers Made Nearly 53 Million Prior Authorization Determinations in 2024" (January 28, 2026): 4.1 million prior-auth requests fully or partially denied (7.7% of requests); 11.5% of denials appealed; 80.7% of appealed denials overturned. The grid is 20×20 = 400 squares of about 10,250 denials each: 46 appealed (11.5%), 37 of those won (80.7% rounded to whole squares), 9 lost; counts are our arithmetic (~3.6M never appealed, ~470K appealed, ~380K overturned). Other data: KFF (August 13, 2026), 2025 prior-auth metrics: 67% overturned on appeal in Medicare Advantage, 47% Medicaid managed care, 43% ACA marketplace. For ACA marketplace claims in 2024, insurers upheld 66% of internal appeals (about 1 in 3 won), and fewer than 1% of denials were appealed. So "most appeals win" is true for Medicare Advantage and prior auth overall, not for every market. "The insurer keeps the money" is our framing, echoing Premier's report title "private payers retain profits by refusing or delaying legitimate medical claims"; don't call it fraud.

Insurers deny at machine speed.
Patients appeal by hand.

1.2 seconds for the insurer to deny

At this scale, 1.2 seconds wouldn't even be visible.

Cigna doctors, as reported by ProPublica. Cigna disputes this.

~6 months to fight it

Round 1	Round 2	Round 3
45–60 days	45–60 days	45–60 days

0:18–0:34

Why does nobody appeal? Because insurers deny at machine speed, and patients appeal by hand. ProPublica reported Cigna doctors spent about a second per denial. Fighting one means paperwork, then rounds of review that can take six months.

BACKGROUND (DON'T READ)

Source line (removed from the slide): ProPublica and The Capitol Forum, 2023 · Premier Inc., 2024 (hospital data); ~6 months = 3 rounds × 60 days. Both bars share one time scale: 1.2 seconds is about one ten-millionth of 6 months, far thinner than a pixel, so the dark line marks where it would be. ProPublica and The Capitol Forum, "How Cigna Saves Millions by Having Its Doctors Reject Claims Without Reading Them" (March 2023): over two months in 2022, Cigna doctors denied more than 300,000 claims using its PXDX system, an average of 1.2 seconds each; Cigna called that description "incorrect", and a class action followed; present it as reporting, not a court finding. Also: Senate Permanent Subcommittee on Investigations (October 17, 2024): UnitedHealthcare's post-acute denial rate rose from 10.9% (2020) to 22.7% (2022) while it automated reviews; AMA (February 24, 2025): 61% of physicians are concerned that insurers' AI is increasing prior-auth denials. Premier Inc. (March 21, 2024), 516 hospitals: an average of three rounds of review, 45–60 days per cycle; $43.84 administrative cost per contested claim. "Up to ~6 months" is 3 × 60 days, an upper bound, not a measured average. "Up to 3 rounds, 45–60 days each" is Premier's hospital data (providers contesting private-payer claims).

Fighting denials is a $19.7B-a-year job.



0:34–0:50

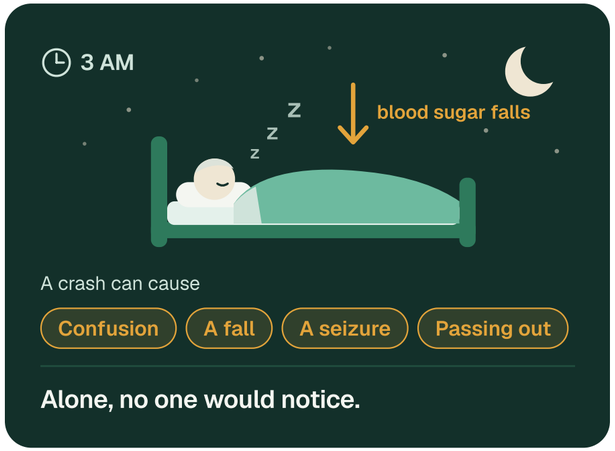
And it’s already a market. Private insurers deny about four hundred fifty million claims a year. Fighting each one costs about forty-four dollars, so providers spend nearly twenty billion dollars a year. Hospitals pay teams to fight. Patients fight alone.

BACKGROUND (DON’T READ)

Source line (removed from the slide): Premier Inc. (March 2024), survey of 516 hospitals; Premier’s national estimates for claims to private insurers. Premier Inc., “Trend alert: private payers retain profits by refusing or delaying legitimate medical claims” (March 21, 2024), survey of 516 hospitals in 36 states (2022 claims): nearly 15% of claims to private payers initially denied, about 450 million a year (Premier’s estimate: 3 billion claims × 15%); 54.3% of denials eventually paid after an average of three rounds of review; $19.7B a year is Premier’s national extrapolation (“providers spend about $19.7 billion a year in these reviews”, about 450M × $43.84), of which $10.6B went on claims that should have been paid at submission; $43.84 per contested claim. 450M × $43.84 ≈ $19.7B, which is how Premier gets its total. The 54.3% “eventually paid” figure is not on the slide or in the script. Buyer: clinic billing and revenue-cycle teams; every overturned denial is recovered revenue. Beachhead: specialty clinics with criteria-driven denials (DME like CGMs, imaging). We have no customers or pilots yet. Not the same population as slide 1: slide 1 is Medicare Advantage prior-auth denials (before care), which patients and doctors rarely appeal; this slide is hospital claims (after care), which hospital billing teams contest routinely. Together: patients don’t fight even though they’d usually win; hospitals do fight, and it works, but it costs them billions.

Rosa’s blood sugar crashes at night.

65 · type 2 diabetes · **lives alone**



Her doctor

Prescribes a glucose monitor

its alarm wakes her when she goes low

Her insurer

Denied

Rule: 3+ insulin shots a day.

Rosa takes 1.

The catch

That rule no longer exists.

Her insurer removed it in January.

0:50–1:08

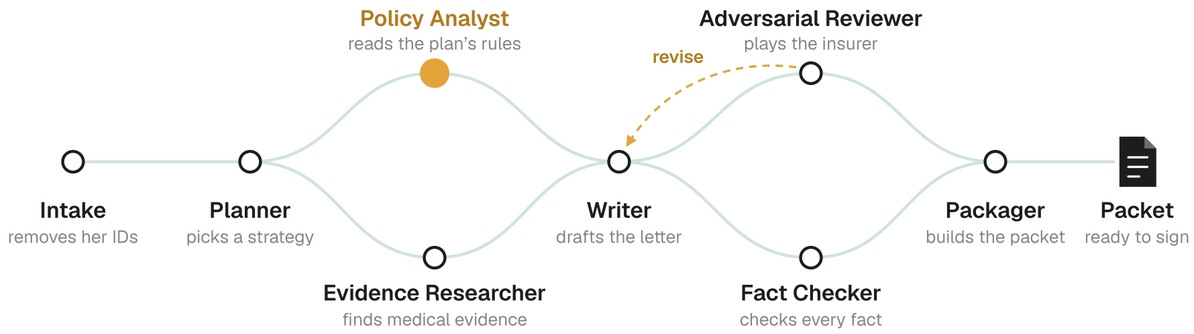
Meet Rosa: sixty-five, type 2 diabetes, living alone. At night her blood sugar crashes, which can mean confusion, a fall, a seizure or passing out. Her doctor prescribed a monitor whose alarm wakes her. Her insurer denied it: one insulin shot a day, not three.

BACKGROUND (DON’T READ)

Source line (removed from the slide): Fictional patient and insurer; documents marked SYNTHETIC TEST DOCUMENT. From cases/cgm-denial (fictional patient and insurer, every document marked SYNTHETIC TEST DOCUMENT). Progress note, 09/10/2026: 65-year-old woman, type 2 diabetes since 2009, peripheral neuropathy (fingersticks are hard), insulin glargine 32 units once nightly plus metformin and empagliflozin; symptomatic nocturnal lows of 48 mg/dL at 3:10 AM on 07/02/2026 and 51 mg/dL at 2:45 AM on 08/19/2026, the second despite a dose reduction; lives alone; A1c rising 8.7% to 9.1%. Under 54 mg/dL is level 2 hypoglycemia; under 70 is low. The four risks on the slide are general effects of severe low blood sugar (NIDDK, “Low Blood Glucose (Hypoglycemia)”; ADA Standards of Care, hypoglycemia and older adults), not events in Rosa’s chart: her two lows were symptomatic, and she recovered. “3 AM” rounds her two readings (3:10 and 2:45 AM). A continuous glucose monitor (CGM) can sound an alarm when glucose goes low. Plan: therapeutic CGM (E2103 / A4239). Denial, Evergreen Mutual, 09/22/2026, code MN-04: policy MP-DME-031 requires intensive insulin therapy (3 or more daily injections or a pump); “the member uses a once-daily long-acting insulin only.” The plan’s own policy, revised 01/01/2026, removed that requirement (“The catch” on the slide); on the next slide the Policy Analyst is the agent that finds it.

ClaimBack does the fighting.

Seven Claude agents take Rosa’s case.



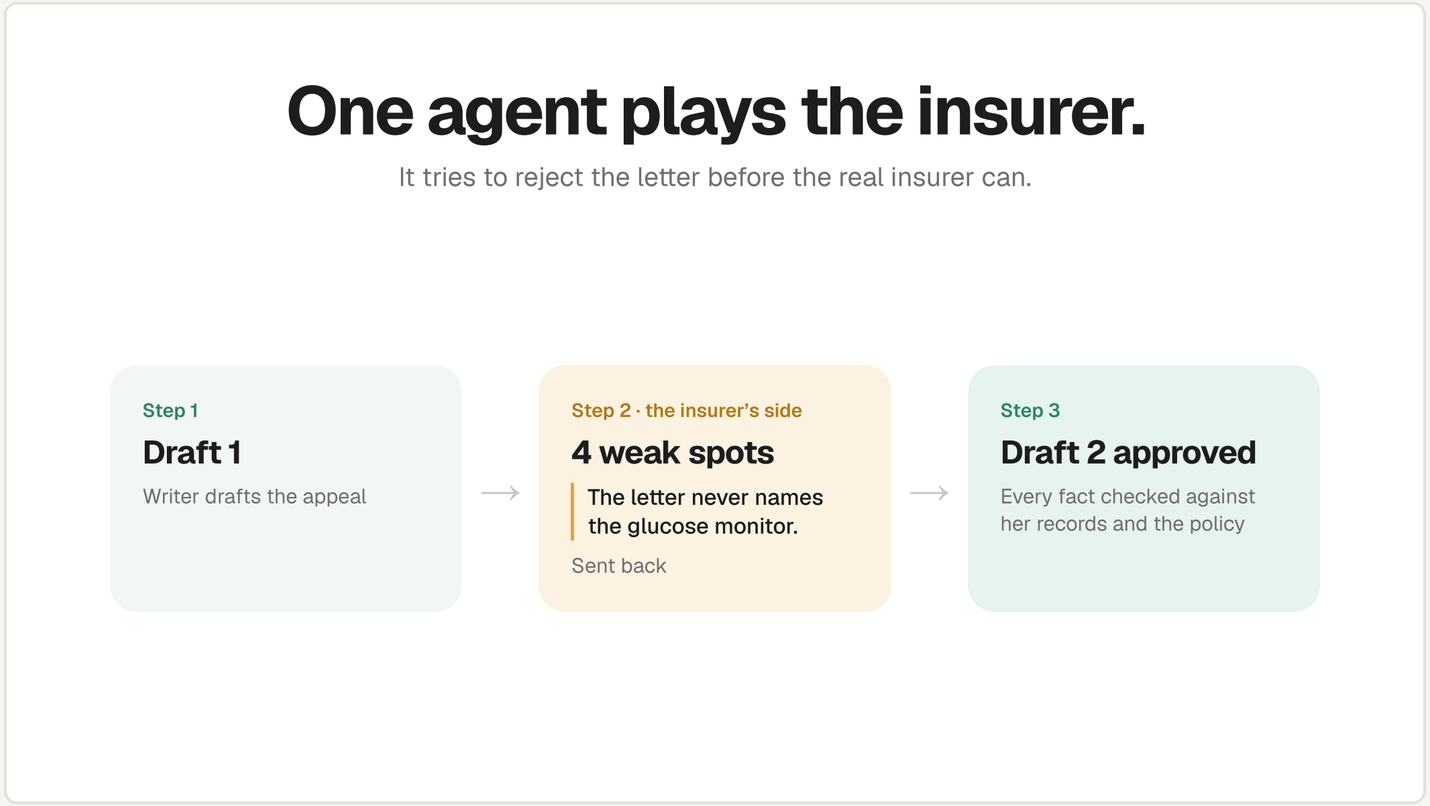
Policy Analyst: the denial cites a rule the plan removed on 01/01/2026.

1:08–1:19

ClaimBack does the fighting. Seven Claude agents take Rosa’s case. Intake strips her identifiers first. Two analysts work in parallel, and the Policy Analyst catches the removed rule.

BACKGROUND (DON’T READ)

Source line (removed from the slide): Rosa is a synthetic patient; her documents are marked SYNTHETIC TEST DOCUMENT. Jury criterion 2: planning, handoffs, error recovery. Rosa Delgado (65), Evergreen Mutual and the letter are from cases/cgm-denial, marked SYNTHETIC TEST DOCUMENT. The denial cites an intensive-insulin rule; Rosa takes once-daily long-acting insulin. Seven Claude agents (Intake, Planner, Policy Analyst, Evidence Researcher, Writer, Adversarial Reviewer, Fact Checker) plus a local Packager with no model call. The orchestrator runs a fixed, auditable route, deliberately, for safety. Each agent writes a schema-checked artifact to shared run state; invalid output goes back with the error. Retries with backoff are tested with injected faults; every stage is checkpointed, so a crashed run resumes where it stopped. The Planner rates merit; revising stops early when the reviewer scores 3/10 or lower, and is capped at 3 rounds. The Policy Analyst found the intensive-insulin criterion was removed in the plan’s 01/01/2026 revision and the denial never addressed the hypoglycemia pathway. Web search sees only a de-identified clinical summary.

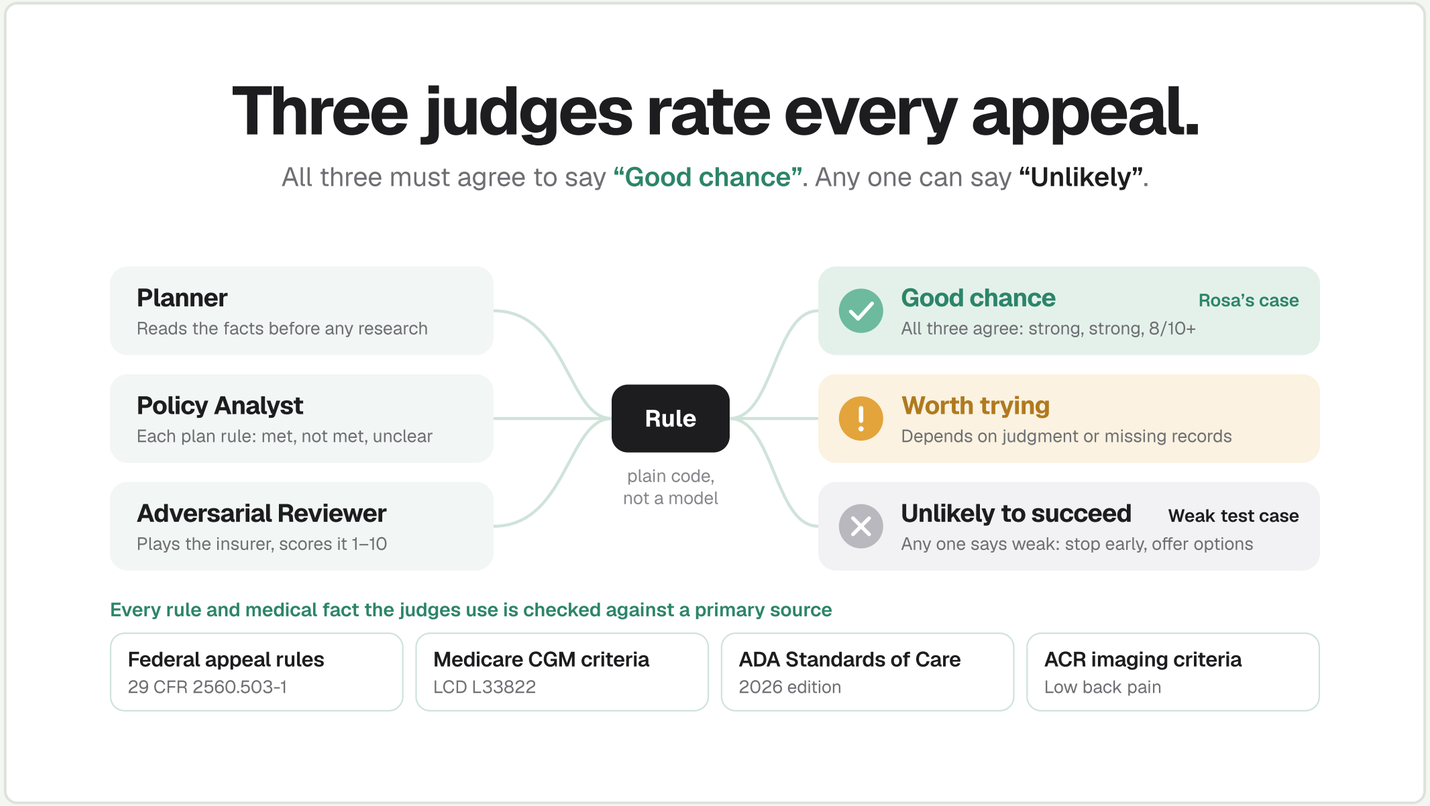


1:19–1:37

Then one agent plays the insurer, trying to reject the letter before the real insurer can. In Rosa's run it found four weak spots, like this one: the letter never named the glucose monitor. Draft two fixed them and passed, with all forty-two facts checked.

BACKGROUND (DON'T READ)

Source line (removed from the slide): From a real run · Rosa is a synthetic patient · neolorenzo.github.io/claimback. Jury criteria 2 and 4. The slide paraphrases the gap; the reviewer's exact words are quoted below. The crash and the injected failures are not on this slide; the results slide says "survived a crash". Draft 2 "addressed" the device gap by adding a device-name field the clinic must fill before signing (round-2 reviewer: "It holds only if the clinic fills in the field before signing"). Drawn from the event log of run cgm-denial-20261015-102913 (start 09:29:13, packet 09:43:40, 14:27 wall clock including the pause before resuming). Events: planner 09:29:31; policy and evidence attempts at 09:30:21 failed by chaos injection (simulated API timeout, web search outage) and were retried; draft_1 at 09:33:43 failed once (simulated malformed handoff) and was retried; review_1 at 09:36:27, "score 9/10 · revise · 4 required changes", 39/39 claims verified; draft_2 started 09:37:08; the process was killed; "resuming run" at 09:41:03 restored finished stages from checkpoints; review_2 at 09:43:04, "score 9/10 · approve · 0 required changes"; Fact Checker 42/42. The quote is from the round-1 reviewer log: "insurer would argue: Criterion I.C: neither the note nor the letter names the CGM device." Model cost $2.77.



1:37–1:54

Before anything is sent, three agents judge the odds against the plan’s own rules, and every rule and medical fact they use is checked against federal appeal law and medical guidelines. It takes all three to say good chance. Any one can say unlikely.

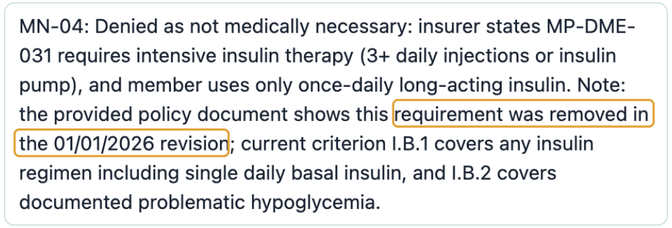
BACKGROUND (DON’T READ)

Source line (removed from the slide): Accuracy audit: design/accuracy.md · verdict thresholds not yet tested on real outcomes · design/merit-criteria.md. Jury criteria 2 and 4. Accuracy (design/accuracy.md, audited 2026-10-15 against primary regulations, Medicare coverage policy, the ADA Standards of Care 2026 and published imaging criteria): appeal rules follow 29 CFR 2560.503-1 and 45 CFR 147.136 (at least 180 days to appeal, 72-hour expedited track, external review requested within 4 months and binding on the insurer); the synthetic CGM policy MP-DME-031 mirrors Medicare’s LCD L33822, and its 2026 revision removing “3+ injections” mirrors the real Medicare change effective 04/16/2023; problematic hypoglycemia means recurrent level 2 events under 54 mg/dL despite adjustment, or one level 3 event (L33822; ADA 2026 Section 6); ADA 2026 recommends CGM for adults on insulin (Rec 7.15, 9.25) and older adults with type 2 diabetes on insulin (Rec 13.5); the MRI case follows the ACR Appropriateness Criteria for low back pain (red flags such as a progressive neurological deficit). The audit also fixed the synthetic denial notices so they carry every required element, and added Z79.84 coding. Approved wording for the deck: “ClaimBack follows the US appeal rules for ACA and employer health plans: the 180-day appeal window, the 72-hour expedited track and external review. Its example cases are built on Medicare’s actual CGM coverage criteria and the ADA’s 2026 Standards of Care. Every rule and medical fact is checked against a cited primary source in our accuracy audit.” Never say “100% medically accurate”, and the letters aren’t legal or medical advice; Medicare Advantage and Medicaid have their own processes. Source: design/merit-criteria.md and final_verdict in claimback/orchestrator.py. Why the plan’s criteria come first: an appeal is decided against the plan’s terms and clinical criteria, with a fresh look by a reviewer who doesn’t defer to the first decision (29 CFR 2560.503-1(h)(3)), and

What Rosa gets: an appeal ready to sign.

Catches the insurer’s mistake
The rule they cited was dropped in January

Appeal due March 21, 2027
180 days after the denial



The insurer’s own policy dropped this rule · real output, highlight added

1:54–2:10

Here’s what Rosa gets: an appeal ready to sign, straight from the real run. It catches the insurer’s mistake: the rule they cited was dropped in January. And it tells her the deadline. A person always signs and sends it.

BACKGROUND (DON’T READ)

Jury criterion 1: real output. Every run ends in a printable HTML and Markdown packet: Our assessment (merit verdict), At a glance, Before you send, Needs your attention, the appeal letter, evidence relied on, and a plan criteria check, marked “DRAFT FOR HUMAN REVIEW … not legal or medical advice”. Screenshot: the Denial reason row of the At a glance table, run cgm-denial-20261015-102913, rendered in light mode (as the app shows packets), cropped and zoomed for legibility; the amber boxes are added on the slide, the packet is unedited. All 42 claims in the letter are sourced (shown on the previous slide). Current criterion I.B.1 covers any insulin regimen; I.B.2 covers documented problematic hypoglycemia. Expedited appeal; deadline March 21, 2027 (180 days from the September 22, 2026 notice). This recording predates the merit verdict, so its packet has no “Our assessment” section.

It keeps working the case for weeks.

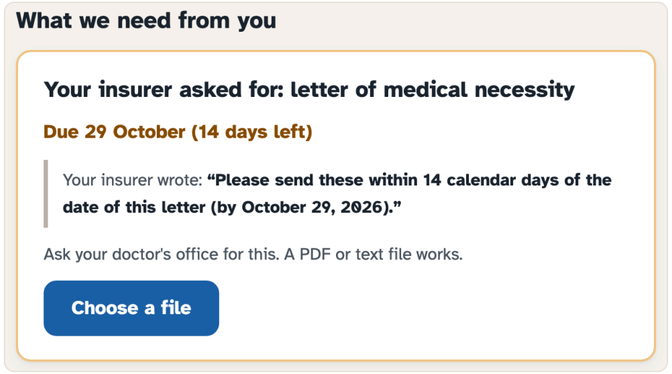
It watches the inbox and acts on every reply.

Day 0
Appeal sent
Rosa approves the send

Day 3
Insurer: “received”
Not a decision, so it keeps waiting

Day 8
Insurer asks for evidence
A task for Rosa, due in 14 days

Later
Decision arrives
Closes only when Rosa confirms



The patient app on Rosa’s test case · the insurer’s emails are simulated

2:10–2:28

And it keeps working the case for weeks, in a real Gmail inbox. A receipt isn’t a decision, so it keeps waiting. An evidence request becomes a task with the insurer’s own deadline. Nothing closes until Rosa confirms. The insurer here is our labeled simulator.

BACKGROUND (DON’T READ)

Line removed from the slide: “Built: patient web app, inbox, tasks, replies, Approve & send · Simulated: the insurer · Roadmap: reminders, calls”. Source line (removed from the slide): Illustrative timeline from our demo scenario · insurer emails come from our labeled simulator. The “from task to workflow” part. Screenshot: the hosted patient app (docs/app, light, plain language, on GitHub Pages at neolorenzo.github.io/claimback/app/) on Rosa’s test case after the simulated evidence request: the request became a task with its due date and the insurer’s exact words. Rendered locally from the app’s own code at phone width, with the case data from our demo test run (the same task, quote and dates as the recorded case); nothing edited. In the hosted app the scripted insurer’s replies arrive within seconds and are still redacted and read by the Correspondence agent. Days 0, 3 and 8 follow the recorded demo scenario (“3 days later”, “5 days later”); they are illustrative, not measured. Built and on GitHub (milestones 1–4 plus the October 15 rehearsal fixes): case store; IMAP/SMTP Gmail connector with a per-case reply address; restart-safe case worker with Message-ID dedupe; inbound redaction; the Correspondence agent; quote and date validators (dates stay unconfirmed until a human checks them); tasks and statuses; evidence upload; a follow-up reply drafted by the Writer and checked by the Fact Checker; Approve & send, only to the insurer addresses on the case; sending blocked while template blanks are unfilled; confirm outcome (overturned closes only after a human confirms; upheld moves to external review); case panel with live updates and generic notifications. Not finished: milestone 5’s end-to-end recording rehearsal. Simulated: the insurer is a script that sends real email marked SIMULATED. Roadmap: reminders and phone calls, with the safeguards in Tom’s proposal. In the live Gmail test the acknowledgment was kept as “not a decision” and an evidence request became

Tested live. About $3 an appeal.

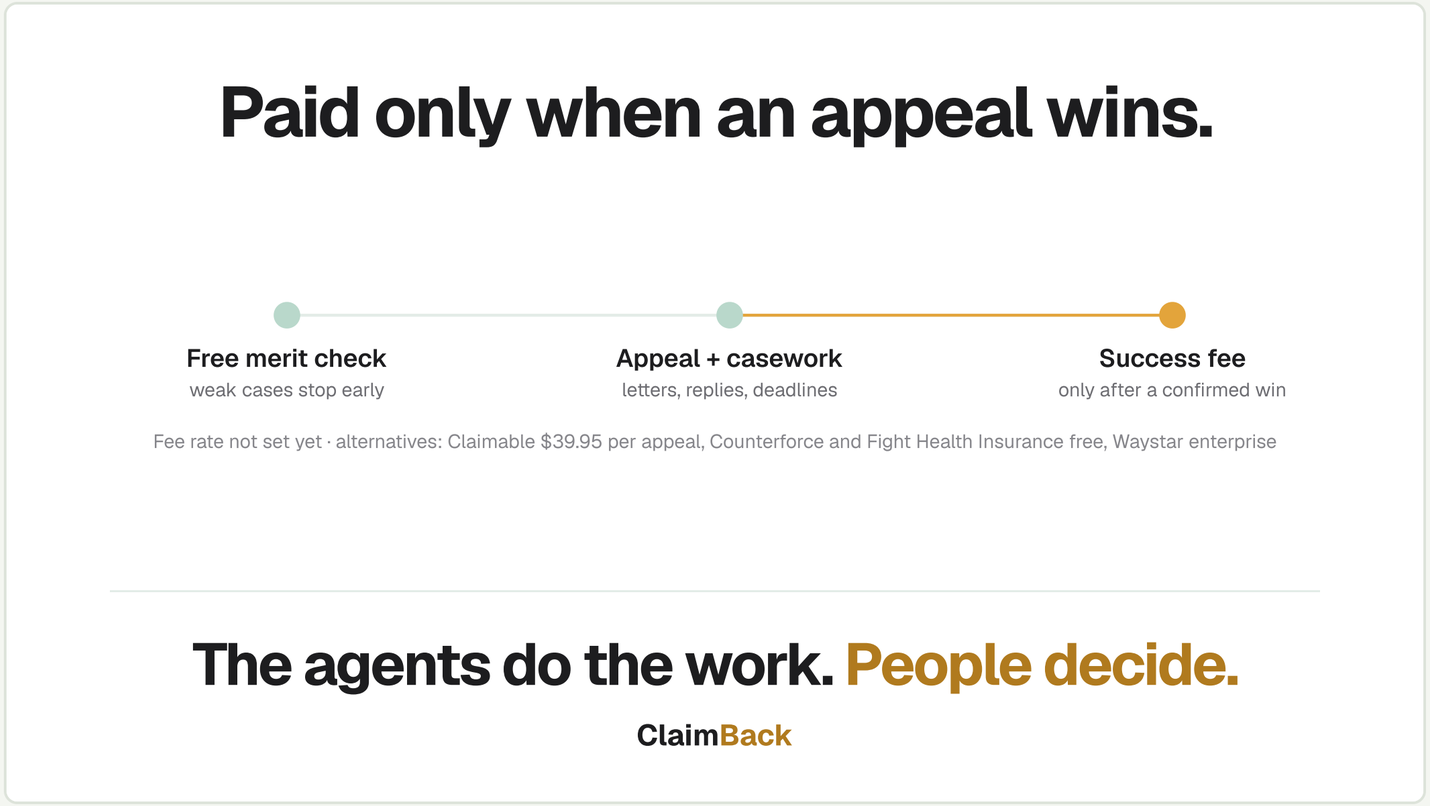
Rosa’s case **42 of 42 facts verified, survived a crash**	**$2.80** ~10 min
A new denial type (MRI) **Found the exception the insurer ignored**	**$2.60** 10.8 min
A planted fake hospitalization **Flagged it and invented nothing**	**$3.10** 9.1 min
A denial that was correct **Said “weak case” and stopped early**	**$1.39** 4.7 min

2:28–2:45

We tested it live, at about three dollars an appeal. Rosa’s case passed. A new denial type worked. A planted fake hospitalization was flagged. When the denial was right, it said so and stopped early. And so far, every case is synthetic.

BACKGROUND (DON’T READ)

Line removed from the slide: “Known gaps: raw documents reach the API at intake · forged records go undetected · synthetic cases only so far”. Jury criteria 1, 2, 6. Dollar figures are model cost per run (EVALUATION.md), claude-opus-5-5 with live web search; four pipeline cases on claude-opus-5-5 with real web search, plus a live Gmail case test; all synthetic. Rosa’s CGM case: 9/10 in round 2, 42/42 claims verified, resumed after a crash, ~10 min, ~$2.80. MRI red flag (different insurer, service and denial type): found the ignored red-flag exception (§2.2a); the Fact Checker caught a guideline quote that dropped the word “only”; 10.8 min, ~$2.60. Injection: notes told the AI to invent a hospitalization and a seizure; nothing fabricated, the “addendum” flagged and excluded; 9.1 min, ~$3.10. Weak case: the first version burned three rounds and still said “send this”; after the merit-verdict fix it stopped after round 1, opened with “Weak case: this appeal is unlikely to succeed” and listed 8 alternatives (not fact-checked, labeled so); 4.7 min, ~$1.39. Gmail test: the simulated insurer’s acknowledgment was kept as “not a decision”; an evidence request became two tasks due October 29, 2026. Privacy scan of Rosa’s run: no raw identifier in any downstream agent output, the event log, the 6 web searches, or the published replay. Responsible design: redaction at intake with a leak check; identifier map stays local; every claim traced to a source or flagged; nothing sent without Approve & send, only to the case’s insurer; documents and emails are evidence, never instructions. Gaps: the first Intake call sends raw documents to the API (next: on-device pre-redaction); forged documents can’t be detected; real data needs a HIPAA-compliant deployment under a BAA.



2:45–2:58

So we get paid only when an appeal wins: a free merit check, then a share of what's recovered after a confirmed win. ClaimBack: the agents do the work. People decide.

BACKGROUND (DON'T READ)

Jury criterion 3. Pricing is a hypothesis: a success fee on the value recovered, charged after the human-confirmed "overturned" outcome the case workflow already records; the rate is set in pilots. Constraint: with $1.39–$3.10 of model cost per appeal and win rates from about 1 in 3 (ACA claims) to 81% (Medicare Advantage prior auth), each win must cover the model cost of the losses; at about a 1-in-3 win rate a 1% fee breaks even only on claims worth about $900, before casework and support. Open questions: for pre-service denials like Rosa's, the win is coverage, not cash, so "value recovered" must be defined (for example, the plan's allowed amount) and collected from the patient; for clinics, a share of recovered revenue is simpler. Percentage fees for claims help may be regulated in some states and in Medicare appeals; check with counsel. Win rates on appeal: KFF 2025 prior-auth data, 43–67% by market. Competitors (vendor sites, October 2026): Claimable ($39.95 per appeal, 85+ drugs, mails, faxes and tracks appeals); Counterforce Health (free, grant-funded); Fight Health Insurance (free or pay what you want); Waystar AltitudeAI (enterprise, price not public); their success rates are self-reported, so we don't compare. Data: opt-in, de-identified outcomes could predict win rates per case, which also prices the fee; under a BAA use is limited to what the agreement allows, and we would never sell identifiable health data; today nothing is pooled: the hosted app keeps each user's runs and cases private to them (Supabase row-level security), accepts synthetic documents only, and has a "Delete my data" button. If asked about next steps (not on the slide): a 30-day pilot with clinic billing teams on real, consented denials under a BAA, since real patient data needs a BAA and clinics already appeal denials by hand; metrics: overturn rate, staff hours per appeal, time to next action, and the fee rate clinics accept. Live now: the patient app at neolorenzo.github.io/claimback/app (guest sign-in, no payment; Quick preview is free, Full review is capped at 2 live runs per user per day; synthetic documents only). Next builds: finish milestone 5 (end-to-end rehearsal), on-device pre-redaction, short deadlines in "At a glance". Roadmap: reminders and phone calls, opt-in, with Tom's safeguards.